Analyze the performance of Machine Learning algorithm to predict Liver cancer

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***Abstract*— The prevalence of lifestyle-related health issues has reached alarming levels, with poor diet, sedentary habits, and environmental factors contributing to the rising incidence of chronic diseases. Liver cancer, in particular, poses a significant threat due to its strong association with unhealthy eating habits, smoking and excessive drinking. Early detection is crucial for effective treatment, yet it remains a daunting challenge for medical professionals despite advances in technology. This paper investigates the application of modern machine learning techniques for the early detection of liver cancer with high accuracy and precision. By leveraging these innovative methods, this paper aim to improve diagnostic outcomes and enhance patient care.**

Keywords*— Machine learning, Liver cancer, Bayes theorem, Decision Tree, Linear regression.

I. INTRODUCTION (*HEADING 1*)

Maintaining a healthy lifestyle is crucial, as every organ plays a vital role in human well-being. Unfortunately, the prevalence of life-threatening diseases is on the rise, and liver cancer is one of the most menacing. The liver is a vital organ liable for food digestion, fat metabolism, and waste management. However, unhealthy habits such as consuming junk food and excessive alcohol consumption can lead to liver cancer, also known as hepatocellular carcinoma. It's a condition characterized by the uncontrolled growth of abnormal cells in the liver. Common symptoms include fatigue, weight loss, itching, and bloating

The global mortality rate due to liver cancer is alarmingly increasing. Historically, diagnosing liver disease was challenging due to limited technological advancements. Fortunately, machine learning has materialized as a ground-breaking technology in the medical field, enabling early cancer detection and supporting medical experts in treating patients. This article explores the application of machine learning algorithms, including Bayes' theorem, decision trees, and linear regression, for predicting liver disease. These algorithms offer promising solutions for early prediction and diagnosis, ultimately improving patient outcomes.

Bayes' theorem leverages probability to predict liver disease outcomes. By analyzing the probabilistic results, the likelihood of liver disease can be accurately forecasted. Decision trees are employed by partitioning the dataset into subsets, which serve as inputs for the tree. The outcomes of each branch are evaluated, enabling the prediction of liver disease. Linear regression is utilized to predict dependent variable values based on independent variables. This algorithm demonstrates potential in envisage about liver disease, leveraging a dataset of previously diagnosed liver cancer cases. This article provides a relative study of various machine learning algorithms, review their performance in predicting liver cancer.

2. LITERATURE SURVEY

Souvik Sarka[1].et.al investigates the use of machine learning (ML) to predict the risk of hepatocellular carcinoma(HCC)in patients with metabolic dysfunction-associated steatotic liver disease(MASLD).The authors emphasize the increasing incidence of HCC among MASLD patients, particularly those without advanced liver fibrosis, who often experience late-stage diagnoses and reduced survival rates. The study utilized data from two institutions, UC Davis and UC San Francisco, to build up and validate a predictive model, achieving an impressive accuracy of 92.06% in identifying HCC risk. The model incorporates standard clinical parameters and aims to serve as a point-of-care tool for healthcare providers, facilitating early risk assessment and targeted screening strategies. though, the authors concede certain restrictions, with the lack of multimodal data and the necessitate for further validation to enhance the model's generalizability. Overall, this research contributes to the evolving role of ML in healthcare, particularly in improving outcomes for patients at risk of liver cancer.

Zi-Mei Zhang [2].et.al suggested that developing machine learning-based predictor for the early diagnosis of hepatocellular carcinoma (HCC), a significant health concern. The researchers utilized gene expression profiles from a large dataset, incorporating both HCC and cirrhosis without HCC (CwoHCC) samples to enhance model training and validation. They employed feature selection methods, including minimum redundancy maximum relevance (mRMR) and maximum relevance maximum distance (MRMD), alongside various machine learning algorithms to create robust predictors. The study identified stable gene pairs and utilized secreted genes to filter features, resulting in several predictors that demonstrated high accuracy and diagnostic capability, with some achieving perfect performance metrics in training datasets. The mRMR + SVM and MRMD + SVM predictors were particularly noted for their effectiveness, achieving high AUC values in validation datasets.

Anita Bakrania[3].et.al narrated the evolving task of artificial intelligence (AI) in the diagnosis and management of liver cancers, which are a significant cause of cancer-related mortality globally. It highlights recent advancements in machine learning (ML) and deep learning (DL) algorithms that aid in pre-screening, diagnosis, and treatment planning for liver cancer patients throughout diagnostic image analysis

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and biomarker discovery. The authors emphasize the need to demystify the "black box" nature of AI to enhance its clinical applicability. Additionally, the paper explores the potential of AI in emerging fields like RNA nano medicine, particularly in streamlining the development of targeted therapies for liver cancer .The authors conclude by discussing future perspectives on AI's integration into liver cancer management, advocating for a multidisciplinary approach to facilitate the transition from research to clinical practice . Overall, the paper presents a complete overview of the current scenery and challenges of AI in liver cancer, underscoring its promise for personalized medicine.

Md Easin Hasan [4].et.al discussed the use of machine-learning classification models to predict hepatocellular carcinoma (HCC) and identify associated genes through explainable AI techniques. The authors implemented various algorithms, including random forest, naïve Bayes, logistic regression, and k- nearest neighbors, achieving high accuracy rates, with random forest reaching 96.53% accuracy and an AUC close to 1.0 .To enhance interpretability, they employed the LIME method, which provides insights into the specific genes influencing predictions for individual patients . The study identified 16 significant genes linked to HCC, which can aid in diagnosis and treatment. The research highlights the importance of understanding gene expression data, especially in the context of high-dimensional datasets, and emphasizes the need for model interpretability to build trust among clinicians and patients . Overall, the proposed framework combines machine learning with explainable AI to improve HCC diagnosis and treatment strategies.
Vishalkumar Vekariya [5].et.al discussed about predicting liver cancer, specifically hepatocellular carcinoma (LIHC), using machine learning techniques applied to epigenomics data. It highlights the significance of epigenetic modifications in cancer development, particularly in liver cells, which are crucial for metabolic functions and regeneration. The study utilizes data from The Cancer Genome Atlas (TCGA), incorporating four types of data: methylation, histone modifications, human genome, and RNA sequences. A total of 1,000 features were initially considered, and various feature selection methods, including Relief F, were employed to refine this to 140 features. The research compares eight different classification methods, ultimately finding that the XGBoost classifier, combined with ReliefF feature selection, achieved the best performance, yielding an area under the curve (AUC) of 1.0 and an accuracy of 99.67% in predicting liver cancer. This work underscores the potential of machine learning in enhancing cancer diagnosis through advanced data analysis techniques.
Ruhul Amin [6].et.al focused on predicting chronic liver disease patients using an integrated approach that combines various projection-based statistical attribute mining methods with machine learning algorithms. It utilizes the Indian liver patient dataset (ILPD) to classify liver disease, addressing issueslike missing values and class disparity in the data. The authors propose a method that integrates techniques such as Principal Component Analysis (PCA), Factor Analysis (FA), and Linear Discriminant Analysis (LDA) to enhance feature extraction and improve classification accuracy. The study incorporates several machine learning algorithms, including logistic regression, random forest, K-nearest neighbor, support vector machine, and multilayer perceptron, achieving an accuracy of 88.10% and an AUC score of 88.20%.The findings suggest that this integrated approach can effectively assist healthcare professionals in diagnosing liver diseases, potentially improving patient outcomes.

III. Dataset Collection

To carry out this research work the numerical value of liver dataset has been considered (https://www.kaggle.com/datasets/uciml/indian-liver-patient-records?resource=download). Prolonged exposure to harmful substances, such as excessive alcohol consumption, spoiled food, and polluted air, can increase the risk of developing liver cancer. This dataset comprises 13 attributes to predict the liver cancer. This proposed work will consider few statistical parameters to predict the disease in an early stage.

IV. Working model

Thefollowingfigure1depictstheprocessofpredictingtheliverca ncerinanearlystage.Thedatasetfrom kaggle repository have been considered for this work. This dataset includes the details of 412livercancer patients and 161 non liver cancer patients

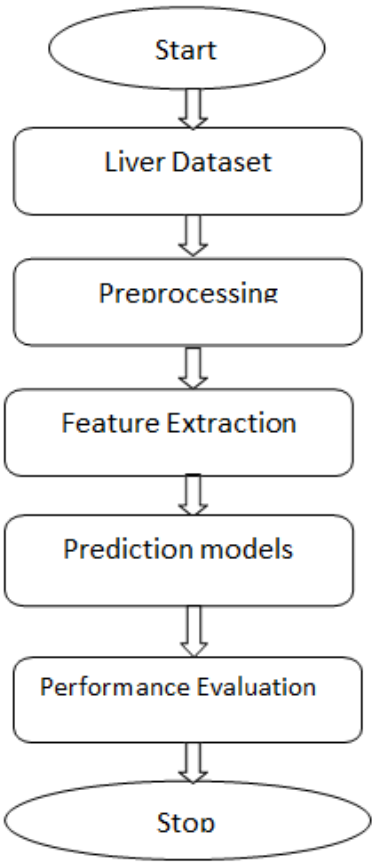

Figure1.Prediction of liver cancer

The Liver dataset will be considered as an input for this research work. Initially it will be involved for preprocessing to remove noisy which is present in the data. From the preprocessed data certain statistical features have been identified such as mean , median, variance and standard deviation to predict the cancer disease. These extracted features will feed into the predictive models comprises Bayes model, Decision Tree and Linear regression to identify the disease in an earlier stage. At the end, the performances of these prediction models have been analyzed with the following parameters precision, Accuracy, Recall and F1 Score.

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ACTUAL VALUES	POSITIVE	NEGATIVE
POSITIVE	TP	FN
NEGATIVE	FP	TN

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F1\,Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

V. Results and Discussion

a. Bayes theorem

It describes the way of occurring an event based on the probability of new evidence. This is also applicable in machine learning algorithm to diagnose the disease. So, in this article have implemented the prediction method using this mechanism.

b. Decision tree:

A Decision tree is like a tree structured model to predict the event by split the data into branches. Each node depicts feature and branches reveals outcomes and final decision will be indicated by leaves. It is also used for classification as well as regression.

c. Linear Regression:

Linear Regression is used for prediction of dependent values based on the independent values. It express the relationship by drawing a straight line using the formula y=mx+c. here the constant c denotes the intercept. This regression model used for predicting the value of y. The following table1 exhibits the performance of various machines learning algorithm to predict the liver cancer.

Table1.Comparison of prediction algorithms

Performance metrics	Bayes Theorem	Decision Tree	Linear Regression
Accuracy	98.71	90.2	92.4
Precision	99.6	95.7	90.2
Recall	99.5	90.3	88.9
F1Score	97.8	92.6	91.3

The above table represents about the performance of predicting the liver disease. According to the table, it has been observed that Bayes theorem have provided an excellent performance compared to all other machine learning algorithms.

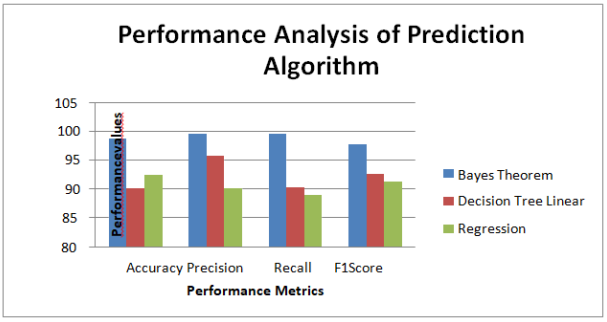

Figure2.Performance Analysis of Prediction Algorithm

The above figure 2 demonstrated about the performance of the few prediction models. The performance of Bayes theorem , Decision Tree and Linear Regression have been compared and analyzed with certain performance metrics. It is noticed that the Bayes theorem have shown better results with 98.71 % of accuracy. Also it has given best performance than decision tree and Linear Regression.

VI. Conclusion

This proposed technique has used liver cancer dataset for the prediction of liver cancer. Although many researchers have suggested various methods to predict of this liver cancer in an early stage, those are not effective due to less precision and accuracy and other parameters. So, this article discussed about the comparison of few machine learning methods with respect to its performance to predict the liver cancer. In this regard, it has been identified that the performance of Bayes theorem have given better results for prediction of liver cancer with 98.71 % accuracy.

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